Text Classification of Movie Dialogue: Comparing Model Performance

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1. Data and Experimental Setup

The provided CSVs contained 0 DARTH VADER lines (the Cornell corpus names him `VADER`), included lines from same-named characters in other films (e.g., the EMPEROR from *Dune*), and the "1000lines" file actually had 552 rows. I therefore re-extracted only the three Star Wars films (filtered by movie ID) from the original Cornell corpus and removed duplicate and label-conflicting lines. The corpus holds only about 490 Star Wars lines for these four characters, so a 1,000-line dataset was not possible.

Dataset	LUKE	LEIA	VADER	EMPEROR	Total	Bad share
Small (Part 1)	94	–	25	6	125	24.8%
Large (Part 2)	229	175	62	14	480	15.8%
Provided Large (comparison)	340	174	0	23	537	4.3%

All configurations share the same repeated stratified 5-fold CV (25 folds). The vectorizer and classifier are wrapped in a Pipeline and fit on training folds only.

2. Consolidated Results

Means over 25 folds; Macro-F1 shown as mean ± standard deviation.

Data	Vectorizer	Model	Imbalance	Acc	Bad P	Bad R	Bad F1	Macro-F1
Small	TF-IDF	LogReg	balanced	0.813	0.660	0.567	0.593	0.735 ± 0.107
Small	SBERT	LogReg	balanced	0.858	0.735	0.708	0.710	0.807 ± 0.088
Large	TF-IDF	LinearSVC	balanced	0.890	0.716	0.537	0.605	0.771 ± 0.035
Large	TF-IDF	LogReg	balanced	0.876	0.622	0.587	0.599	0.763 ± 0.040
Large	SBERT	LogReg	balanced	0.847	0.522	0.722	0.599	0.752 ± 0.046
Large	Count	MultinomialNB	none	0.896	0.826	0.437	0.560	0.751 ± 0.070
Large	Count	GradBoost	balanced	0.856	0.588	0.507	0.528	0.721 ± 0.064
Large	Count	RandomForest	none	0.870	0.888	0.198	0.315	0.622 ± 0.074

3. Research Questions and Final Model

Q1. Data scale: Macro-F1 rose and its SD fell in 19 of 20 configurations (final model SD 0.107 → 0.040), but Bad Recall barely moved (0.567 → 0.587).

Q2. Accuracy vs. Bad Recall: On Large, the top-accuracy model was Count + NB (0.896), but its Bad Recall was only 0.437. Unweighted TF-IDF LR reached 0.842 accuracy while predicting almost everything as Good (Bad Recall 0.003). On the provided Large file, an RF finding zero Bad lines still scored 0.957.

Q3. Count vs. TF-IDF: TF-IDF raised Macro-F1 by 0.03–0.05 for balanced linear models; Count was better for NB and RF. With a median line length of 6 words, counts are mostly 0/1, so L2 normalization likely mattered more than IDF.

Q4. Most balanced model (final choice): TF-IDF + Logistic Regression (balanced): 1st on Small (0.735), 2nd on Large (0.763). Like LinearSVC (Large 1st, 0.771), it keeps Bad Precision and Recall above 0.5; RF never exceeded 0.20 Bad Recall.

Q5. Pre-trained embeddings: SBERT (all-MiniLM-L6-v2) + LR (balanced) beat TF-IDF on Small (Macro-F1 0.807 vs. 0.735). On Large, higher Bad Recall (0.722 vs. 0.587) but lower Precision (0.522) left Macro-F1 similar (0.752). At ~12 ms per line on CPU, it is worth using for small data or when Bad Recall matters.

4. Misclassification Analysis

The final model (Large) missed 33 of 76 Bad lines and flagged 24 Good lines, relying on topic words and the villains' commanding style ("the").

Actual → Predicted	Speaker: Line	Cause (contributing terms)
Bad → Good	VADER: What is thy bidding, My Master?	what (+0.28) offsets my master (−0.41)
Good → Bad	LUKE: Lord Vader … Is the dark side stronger?	topic word dark side (−0.23)
Good → Bad	LEIA: Yes.	yes (−0.74), from Vader's lines

5. Limitations and Improvements

- **Limitations:** 76 Bad lines (~15 per fold) give Bad SDs of 0.06–0.10, too wide to rank top models; "Yes." lines are unclassifiable, and Large changed size and cast together.
- **Improvements:** Nested CV, threshold tuning, more villains, LLM comparison.